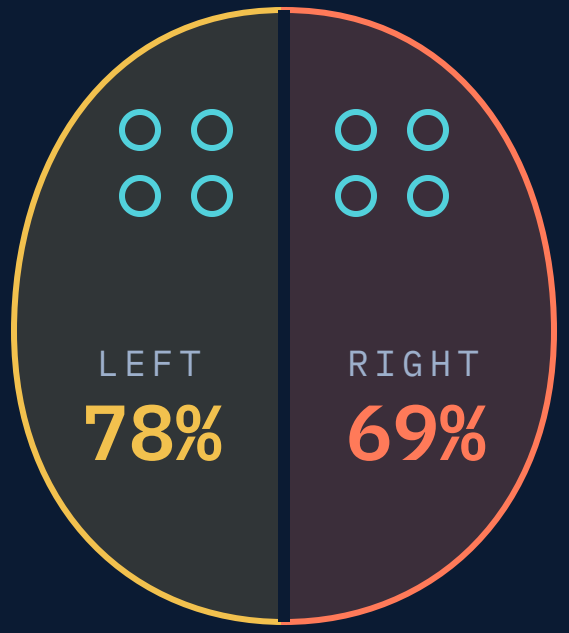
DIGITAL TWIN ENGINEERING · DT-G04

CereFlow-*DT*

A digital twin designed to watch **the brain's blood supply**, live.



FIRST DATA CHECK · PROTOTYPE

69%

In our first look at the public data, one volunteer stood up and flow in the right side of the brain fell to **69% of their baseline** within seconds. *Would today's monitor tell anyone?*

STAND UP

Monitors show waveforms. Clinicians need *answers*.

1

Is enough blood reaching each side of the brain?

Compared with this person's own baseline, not a textbook average.

2

Is the brain still holding its flow steady?

Cerebral autoregulation: does flow just follow blood pressure?

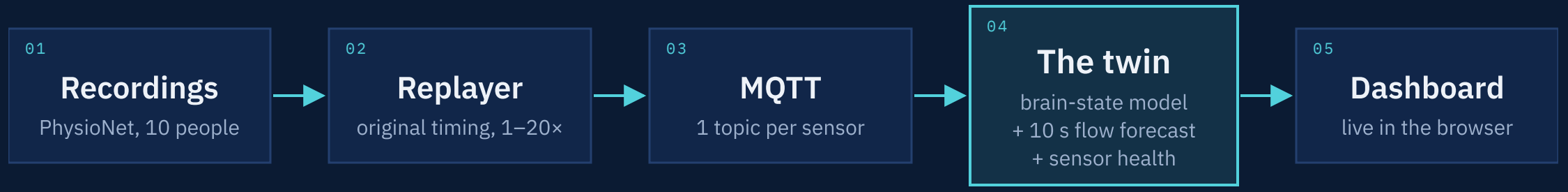
3

Is flow about to drop, and can we trust the sensors?

Early warning, broken-sensor detection, and a model that says when it's unsure.

FOR | Clinicians running stand-up / tilt tests | Neurocritical care after an aneurysm bleed | Engineers tuning the monitoring

Sensor streams in, *live brain state* out



Replayer and twin talk only through the broker, so a live patient feed can later replace the replay with nothing else changing.

52

sensor streams of 4 types to ingest: Doppler, blood pressure, breathing, NIRS

10

healthy adults in the public dataset, ~33 min each: lying, then standing

4

live states the twin tracks: perfusion, autoregulation, posture phase, sensor health

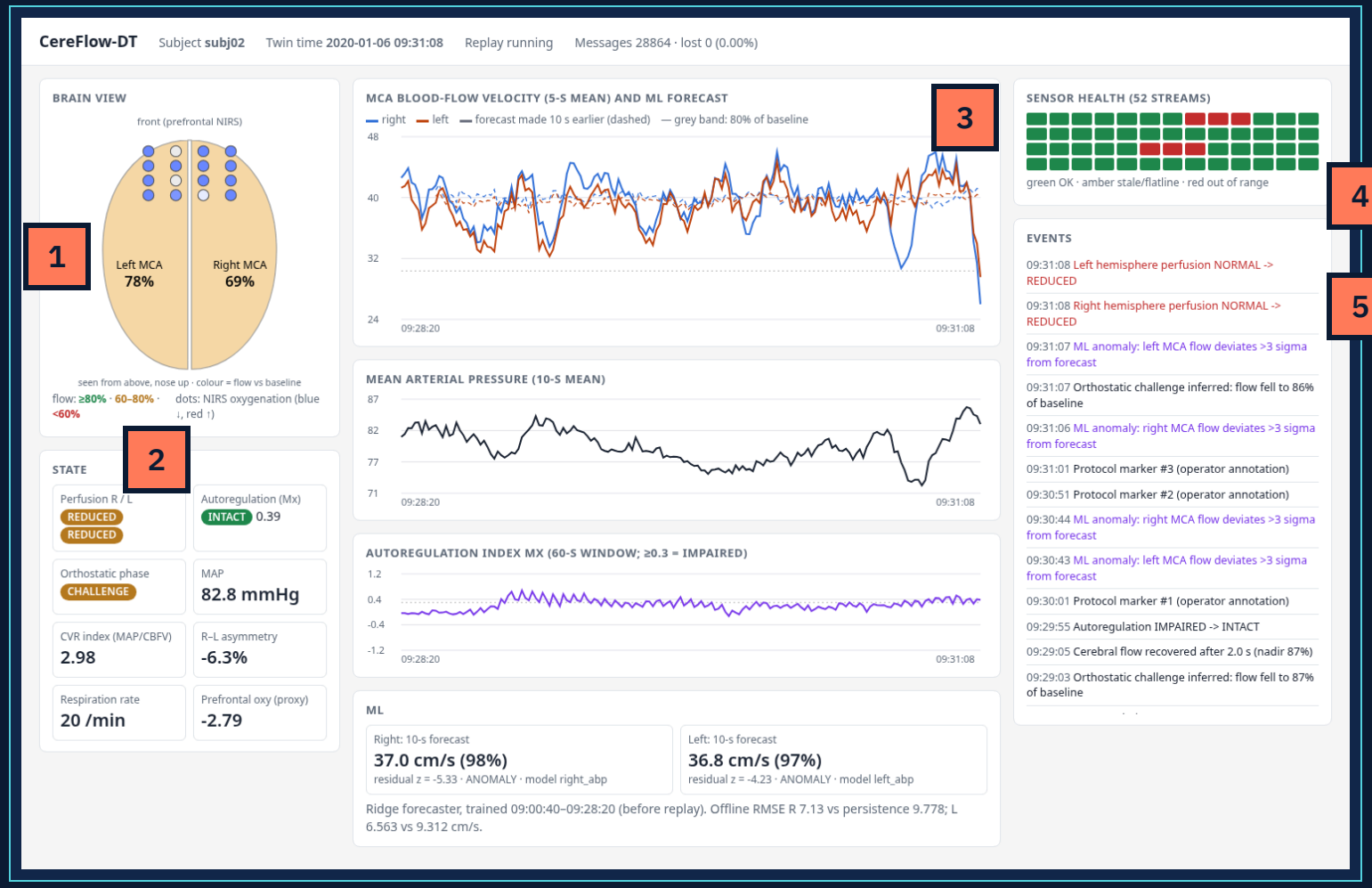
10 s

flow forecast per hemisphere, trained only on data the twin has already seen

DATA · PhysioNet "Cardiorespiratory response to orthostatic stress" v1.0 · Mukli et al., Front. Physiol. 2021

Concept: the moment a patient *stands up*

EARLY PROTOTYPE



1. **Brain map** flow on each side as a % of this person's own baseline.
2. **State** perfusion, posture phase and autoregulation in plain words.
3. **Forecast** a drop the 10-second forecast did not expect is flagged as an anomaly.
4. **Sensor health** channels with bad contact turn red, so a loose sensor isn't read as a brain event.
5. **Event log** every state change, time-stamped.

PROTOTYPE · first check on replayed public data (subject 02), not a validated result

Success criteria, *set before we measure*

Beat

"no change" as a 10 s flow forecast

on most hemispheres, scored on unseen data

0

messages lost when replaying faster than real time

with replay timing kept close to the original

All

known faulty sensors flagged

and a fallback model when blood pressure fails

Seconds

from standing up to a stand-up alert

for every subject, with tuned thresholds

Each criterion becomes an automated check `python verify_report.py` rerun on every change

From design to *bedside twin*

01 Build the full twin on the replayed public data and run every check

02 Tune alarm thresholds on held-out subjects

03 Add a live "coupling network" view of heart, breathing and brain, after Mukli et al. 2021

04 Later: validate on patient data after aneurysm bleeds

ONE ASK

The dataset has elapsed seconds, not dates. We anchor it to a fixed, configurable start time.

Can we confirm this counts as an absolute timestamp?

CereFlow-DT: *know the brain is getting enough blood*, before it isn't.